

MID-SEMESTER EXAMINATION SEPT. 2025

Name of the Program: B. Tech

Semester: III

Name of the Course: Object Oriented Programming with C++

Course Code: TCS-307

Time: 90 Minutes

Maximum Marks: 50

Note:

- Answer all the questions by choosing any one of the sub-questions
- Each question carries 10 marks

Q1	(10 Marks)	
(a)	Create two namespaces: Department_CS and Department_EE. Each namespace should contain a Student class (with id, name, and course) and a Faculty class (with id, name, and subject). Both classes must include a displayInfo() function. In the main function, demonstrate the creation of Student and Faculty objects from both namespaces and show how the scope resolution operator (::) accesses the correct classes/functions from namespaces. Additionally, implement a third namespace, University, containing a common utility function printWelcomeMessage() accessible to all departments.	CO1
	OR	
(b)	Differentiate between a reference variable and a pointer variable in C++. Explain the two different uses of the 'this' pointer in C++.	CO1
Q2	(10 Marks)	
(a)	Design a C++ program to demonstrate string manipulation for a publishing company for editing an article. Use the initial string: "The quick brown fox jumps over the lazy dog". Perform the following string operations step by step: 1) Insert "swift " at position 10. 2) Erase "lazy " starting at position 35. 3) Extract "brown fox" using substr(). 4) Find the position of "swift". 5) Replace "dog" with "cat". Apply insert(), erase(), substr(), find(), and replace() effectively.	CO1
	OR	
(b)	Why are classes preferred over structures in C++ for implementing data hiding? Create a Student structure with roll number, name, and marks. Then convert it into a class where roll number and marks are private, and only the name is public. Write functions to securely set and retrieve roll number and marks.	
Q3	(10 Marks)	
(a)	Consider a Student class with attributes: name and a pointer to marks (dynamically allocated integer). Write a parameterized constructor to initialize the name and allocate memory for marks. Write a copy constructor to copy values. Demonstrate in main() how modifying one object's marks affects the other when: (a) A default shallow copy is used. (b) A user-defined deep copy is implemented.	CO2
	OR	
(b)	1) What is a destructor in C++? Provide an example where a destructor is used to free dynamically allocated memory. 2) Explain the different uses of the 'static' keyword in C++.	
Q4	(10 Marks)	
(a)	Design a class Book in C++ with data members: bookID, title, author, and price. Include member functions input() to accept details and display() to show them. In the	CO2
	main function, use a <i>pointer to an object</i> to dynamically allocate memory for multiple books, call the member functions using the pointer, and finally deallocate the memory.	
	OR	
(b)	What is a friend function in C++? How does it differ from a member function? Create two classes, Rectangle and Circle. Write a friend function compareArea() that compares the areas of both shapes and displays which one is larger.	
Q5	(10 Marks)	CO3
(a)	Differentiate between overloading using member functions and overloading using friend functions with an example.	
	OR	
(b)	Design a class Matrix to represent a 2x2 matrix. Use a friend function to overload the + operator for matrix addition. Display the resulting matrix after adding two matrices.	CO3

